MMRD Nova Distance Calculator

The **MMRD nova distance calculator** estimates the distance to a nova from its rate of decline after maximum light. It is intended to do for novae what the Leavitt's Law distance calculator does for Cepheids: combine a measured light-curve quantity with a published luminosity calibration and the distance modulus.

This plug-in appears in VStar's **Tool** menu as **MMRD nova distance calculator**.

References

The implementation is based primarily on:

- Kantharia, N. G. 2017, *Novae: I. The maximum magnitude relation with decline time (MMRD) and distance*, arXiv:1703.04087.
- Kok, Y. 2010, *Absolute Magnitudes and Distances of Recent Novae*, JAAVSO, 38, 193.

The historical MMRD calibrations are taken from Kok's summary of Schmidt (1957), Cohen (1985), della Valle & Livio (1995), and Downes & Duerbeck (2000), with the sign convention cross-checked against Kantharia (2017).

What the Calculator Does

The maximum magnitude versus rate of decline (MMRD) method estimates the peak absolute magnitude, M_V , from the time taken for a nova to decline by 2 or 3 magnitudes from maximum:

- t_2 : days from maximum to maximum + 2 magnitudes.
- t_3 : days from maximum to maximum + 3 magnitudes.

Once M_V is estimated, the distance is computed from the extinction-corrected distance modulus:

$$D = 10^{0.2(m_V - A_V - M_V + 5)} \text{ pc}$$

where:

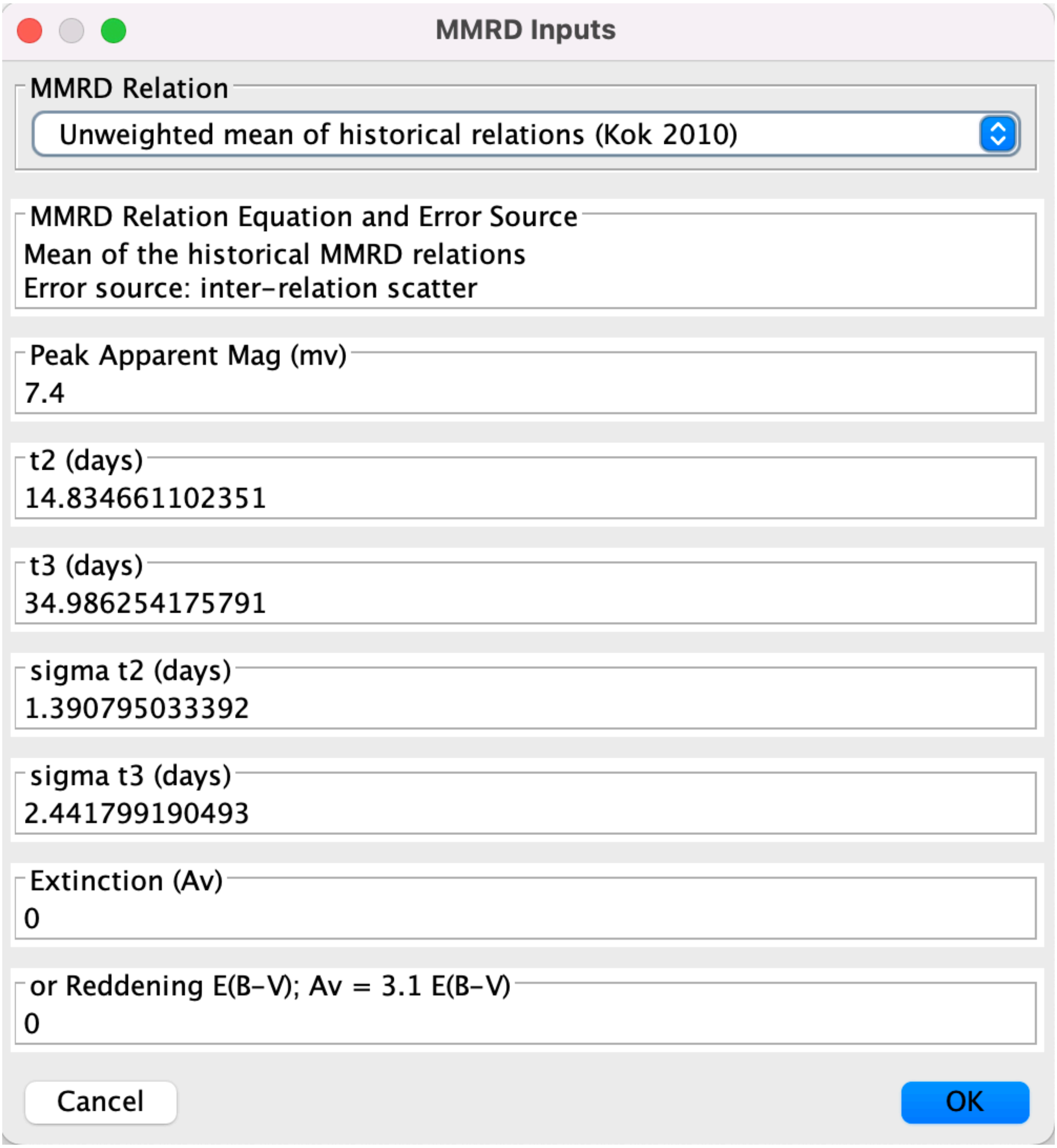
- m_V is the peak apparent magnitude,
- A_V is visual extinction,
- D is distance in parsecs.

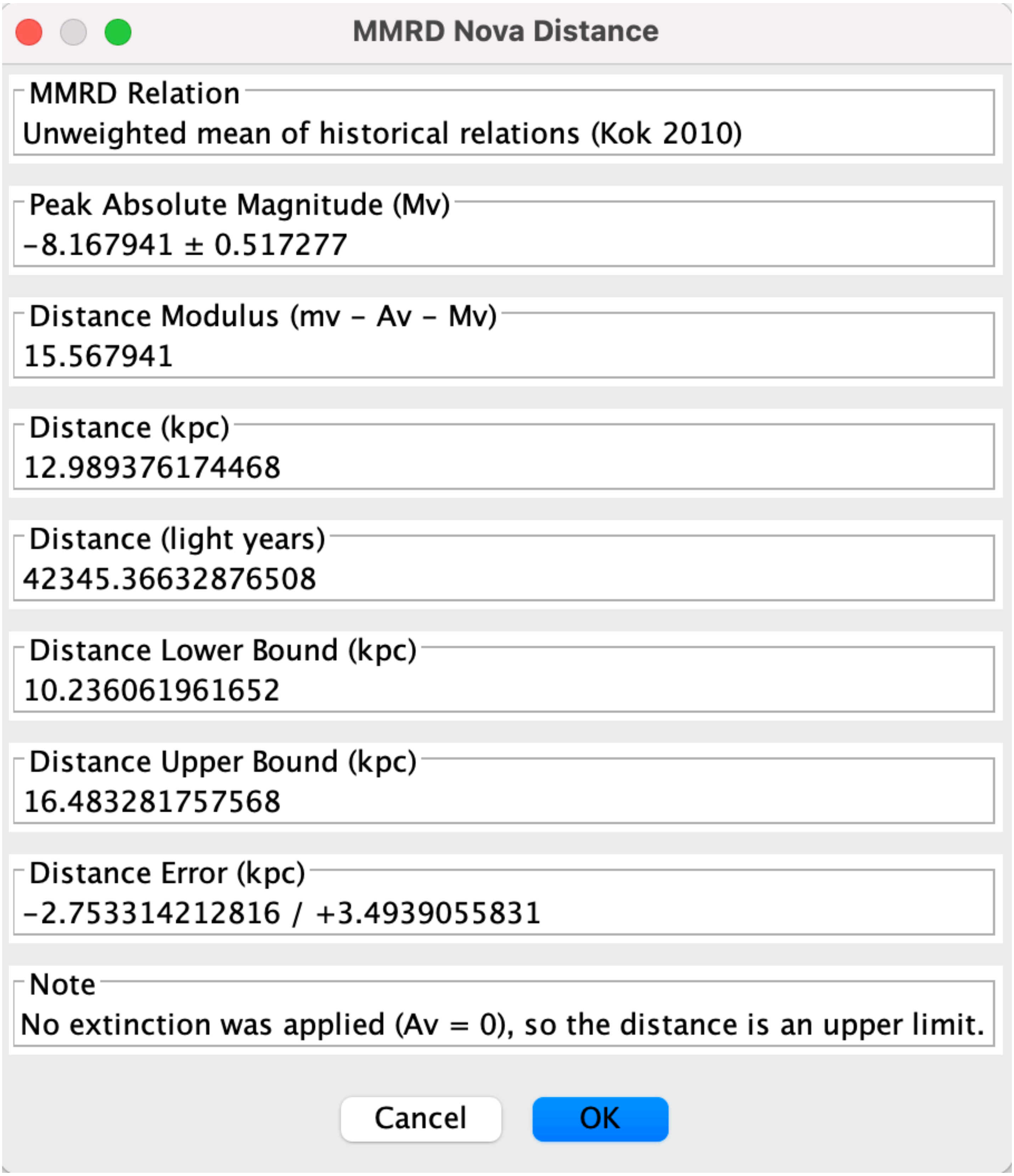
The result dialog reports distance in kiloparsecs (kpc), which is the unit used in the nova distance literature, and includes light years for the point estimate as a convenience.

If $A_V = 0$, the reported distance should be treated as an upper limit unless extinction is known to be negligible.

Basic Workflow

1. Load nova observations, preferably around the outburst and early decline.
2. Choose **Tool > MMRD nova distance calculator**.
3. Select the observation series to use.
4. Choose a source for t_2 and t_3:
    - **Exponential model fit (Kok 2010, eq. 10)** for raw visual or V data.
    - **Directly from observations** only for clean or already-smoothed data.
5. Review or edit m_V, t_2, t_3, optional σt_2 and σt_3, and extinction.
6. Choose an MMRD relation.
7. Read the resulting absolute magnitude, distance, and lower/upper distance bounds.


MMRD Inputs
MMRD Relation
Unweighted mean of historical relations (Kok 2010)
MMRD Relation Equation and Error Source
Mean of the historical MMRD relations
Error source: inter-relation scatter
Peak Apparent Mag (mv)
7.4
t2 (days)
14.834661102351
t3 (days)
34.986254175791
sigma t2 (days)
1.390795033392
sigma t3 (days)
2.441799190493
Extinction (Av)
0
or Reddening E(B–V); Av = 3.1 E(B–V)
0
Cancel
OK



Choosing the JD Range

For comparison with Kok (2010), use the same outburst window shown in the paper's figures. For example, Kok's Figure 4 for V4633 Sgr has an x-axis labelled `JD + 2450894.5000` and spans 0 to 60 days, corresponding to:

$$2450894.5 \le JD \le 2450954.5$$

Using a substantially longer time range can change the exponential fit, the asymptote, and hence t_2 and t_3.

Light-Curve Parameter Sources

Exponential Model Fit

This option fits Kok's equation (10):

$$m(t) = P_1 - P_2 e^{-P_3(t-t_0)}$$

The fit is performed from the brightest observation onward. The plug-in then measures t_2 and t_3 as the fitted curve's crossing times at $m_0 + 2$ and $m_0 + 3$, where m_0 is the observed maximum magnitude.

This is the recommended source for raw AID visual data. Raw observations can scatter above and below a smooth decline, so direct crossing detection may find a too-early crossing.

This is the same fit provided by the standalone **Nova Exponential Decline Model** plug-in; see its documentation for the parameters, decline-time derivation, fit metrics, and uncertainty handling.

Directly from Observations

This option finds the brightest observation and linearly interpolates the first crossings at peak + 2 and peak + 3 magnitudes. It is useful for clean, well-sampled, monotonic data or a precomputed model/means series, but is not recommended for noisy visual data.

MMRD Relations

The default relation is:

$$M_{V,0} = 2.16 \log_{10}(t_2) - 10.804$$

from Kantharia (2017). It is the preferred modern single-relation calibration and remains first in the drop-down menu.

The plug-in also includes historical relations and two aggregate options:

- **Unweighted mean of historical relations (Kok 2010)**: mean of the historical Kok relations; error is the inter-relation scatter. This is the best option for reproducing Kok's Table 3 and Table 4.
- **Inverse-variance weighted mean of historical relations**: weighted mean using the published coefficient errors and optional t_2/t_3 errors. Its reported error is the larger of the formal weighted error and the inter-relation scatter, avoiding over-precise error bars when calibrations disagree.

Individual historical relations are also available for comparison and study.

Table 4 in Kok (2010)

Kok's Table 4 columns are not different MMRD relations. They are different extinction assumptions applied to the same Table 3 absolute magnitude:

- $A_V(1)$ and $D(1)$ use the Schlegel et al. (1998) full line-of-sight extinction.
- $A_V(2)$ and $D(2)$ use the Arenou et al. (1992) three-dimensional extinction model.
- $D(r)$ lists distances from other published work.

To reproduce Table 4, use Kok's Table 1 values for m_0, t_2, and t_3, select **Unweighted mean of historical relations (Kok 2010)**, and enter the appropriate A_V value from Table 4.

Extinction

You can enter either:

- A_V directly, or
- reddening $E(B-V)$, in which case the plug-in uses:

$$A_V = 3.1E(B-V)$$

If both are supplied and A_V is non-zero, A_V takes precedence.

Error Bars and Bounds

The papers use error bars at several stages, and these do not all mean the same thing. The plug-in keeps the stages explicit:

Stage	Meaning
σt_2, σt_3	uncertainty in the decline times
σM_V for a single relation	propagated from published coefficient errors and optional t_2/t_3 errors
σM_V for unweighted mean	scatter among the historical relations
σM_V for inverse-variance mean	max(formal weighted error, inter-relation scatter)
distance bounds	σM_V propagated through the distance modulus

When the **Exponential model fit (Kok 2010, eq. 10)** source is used, the σt_2 and σt_3 fields are filled in automatically by propagating the fit's parameter uncertainties through the closed-form crossing time (see the Nova Exponential Decline Model documentation). If the fit cannot estimate those uncertainties, or the decline times are taken directly from the observations, the fields are left blank; you may enter values manually, for example from a published table.

Distance bounds are shown in kpc as lower and upper bounds rather than a single symmetric error, because distance depends exponentially on magnitude. A symmetric σM_V usually becomes an asymmetric distance interval.

Comparison with Kok (2010)

For each nova:

1. Match the JD window shown in Kok's figure.
2. Compare the fitted t_2 and t_3 with Table 1.
3. Compare M_V with Table 3.
4. Enter Kok's $A_V(1)$ or $A_V(2)$ and compare distance with Table 4.

For V4633 Sgr, using Kok's Table 1 values:

- $m_0 = 7.6$
- $t_2 = 19.5$
- $t_3 = 42.9$

the unweighted historical mean gives $M_V \approx -7.72$, close to Kok's published $M_V = -7.7 \pm 0.3$.

Limitations

- The plug-in does not look up extinction automatically.
- The Kok aggregate is a reconstruction; Kok's paper does not include source code or a fully specified weighting implementation.
- The MMRD method is empirical and calibration-dependent.
- Raw visual data can make peak detection and direct crossing detection fragile; use the exponential fit and inspect the model series.